



DF & ADF Stationarity Test

Dickey-Fuller Test

Definition

- The **Dickey-Fuller (DF) Test** is a statistical hypothesis test used to determine whether a time series contains a **unit root**, i.e., whether the series is **non-stationary**.
- It is designed for the simplest autoregressive model: $AR(1)$

Why do we need these tests?

Before fitting any time series model, it requires stationarity. By using these tests we can answer the question:

| Does the series contain unit root or it is stationary?

If the data contains a unit root:

- Parameter estimates become unreliable.
- Statistical inference becomes invalid.
- Regression may become spurious.

Econometric models which require stationarity: VAR, ARIMA, VECM, Kalman Filter, State-Space Models, Cointegration test

Meaning in finance

Suppose any asset's today's price is X_t

If $X_t = X_{t-1} + \varepsilon_t \rightarrow$ today's price entirely depends on yesterday's price + shock

Shock affect the future prices and the effect of shocks remain permanent. $\rightarrow$ Such processes are **Non-Stationary**

Dickey-Fuller & Augmented Dickey-Fuller test confirms if the time series is following random walk or is it stationary?

Mathematical Derivation

Consider a time series process $AR(1)$:

$$X_t = \phi X_{t-1} + \varepsilon_t$$

where $\varepsilon_t \sim \text{i.i.d. } \mathcal{N}(0, \sigma^2)$

Defining the 1st difference ΔX_t :

$$\begin{aligned}\Delta X_t &= X_t - X_{t-1} \\ \therefore \Delta X_t &= [\phi X_{t-1} + \varepsilon_t] - X_{t-1} \\ \therefore \Delta X_t &= [\phi - 1]X_{t-1} + \varepsilon_t \\ \therefore \boxed{\Delta X_t = \gamma X_{t-1} + \varepsilon_t} \quad (\because \gamma = \phi - 1)\end{aligned}$$

$\Delta X_t = \gamma X_{t-1} + \varepsilon_t$ is **Dickey-Fuller** Regression equation.

Hypothesis

- Null Hypothesis (H_0): If $\gamma = 0 \Leftrightarrow \phi = 1$ (Unit root exist $\rightarrow$ Non Stationary)
- Alternative Hypothesis (H_1): $\gamma < 0 \Leftrightarrow \phi < 1$ (No unit root $\rightarrow$ Stationary)

Decision: If $\gamma = 0 \rightarrow$ Unit root exist $\rightarrow$ Non Stationary, otherwise Stationarity

Regression Variants

1. **No Constant, No Trend (NCNT):** Use when series fluctuate around 0, which is rare for financial prices.

$$\boxed{\Delta X_t = \gamma X_{t-1} + \varepsilon_t}$$

2. **With Constant, No Trend (WCNT):** Use when series has non zero mean but no systematic trend. This variant is most common for log prices.

$$\boxed{\Delta X_t = \alpha_0 + \gamma X_{t-1} + \varepsilon_t}$$

3. **With Constant, Linear Trend (WCLT):** Use when series has visible upward/downward trend.

$$\boxed{\Delta X_t = \alpha_0 + \beta t + \gamma X_{t-1} + \varepsilon_t}$$

Limitations of Dickey-Fuller Test:

The Dickey-Fuller test assumes ε_t is **White Noise**.

However, financial time series usually have

- Autocorrelation
- Serial Dependence
- Higher-Order dynamics

Therefore, if $AR(p)$ becomes higher order, ε_t will be serially correlated. That is why Dickey-Fuller test become **unreliable** on higher order $AR(p)$.

Augmented Dickey-Fuller Test

Definition

The **Augmented Dickey-Fuller Test** extends the Dickey-Fuller test by adding lagged difference terms to remove serial correlation from the residuals.

Mathematical Formulation & Derivation

Consider $AR(p)$ process,

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

where,

- X_t = Value of time series at time t
- $X_{t-1}, X_{t-2}, \dots, X_{t-p}$ = Previous observations. (lag 1, lag 2, lag 3, ..., lag p)
- $\phi_1, \phi_2, \dots, \phi_p$ = Autoregressive Coefficient
- ε_t = White Noise/ Error term
- Subtracting X_{t-1} from both sides, and expressing all lagged levels X_{t-k} for $k \geq 2$

$$X_{t-k} = X_{t-1} - \sum_{j=1}^{k-1} \Delta X_{t-j}$$

$$\begin{aligned}
\therefore X_t - X_{t-1} &= [\phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t] - X_{t-1} \\
\therefore \Delta X_t &= (1 - \phi_1) X_{t-1} + \left[\sum_{k=2}^p \phi_k X_{t-k} \right] + \varepsilon_t \\
\therefore \Delta X_t &= (1 - \phi_1) X_{t-1} + \left[\sum_{k=2}^p \phi_k \left(X_{t-1} - \sum_{j=1}^{k-1} \Delta X_{t-j} \right) \right] + \varepsilon_t \\
\therefore \Delta X_t &= (1 - \phi_1) X_{t-1} + \left[\sum_{k=2}^p \phi_k X_{t-1} \right] - \left[\sum_{k=2}^p \phi_k \left(\sum_{j=1}^{k-1} \Delta X_{t-j} \right) \right] + \varepsilon_t \\
\therefore \Delta X_t &= \left(\sum_{k=1}^p \phi_k - 1 \right) X_{t-1} - \left[\sum_{k=2}^p \phi_k \left(\sum_{j=1}^{k-1} \Delta X_{t-j} \right) \right] + \varepsilon_t
\end{aligned}$$

- Interchange the double summation in 2nd term

$$\begin{aligned}
\sum_{k=2}^p \phi_k \left(\sum_{j=1}^{k-1} \Delta X_{t-j} \right) &= \sum_{j=1}^{p-1} \left(\sum_{k=j+1}^p \phi_k \right) \Delta X_{t-j} \\
\therefore \Delta X_t &= \left(\sum_{k=1}^p \phi_k - 1 \right) X_{t-1} + \left[\sum_{j=1}^{p-1} \left(- \sum_{k=j+1}^p \phi_k \right) \Delta X_{t-j} \right] + \varepsilon_t
\end{aligned}$$

where, $\sum_{k=1}^p \phi_k - 1 = \gamma$ and $\left(- \sum_{k=j+1}^p \phi_k \right) = \delta_j$

$$\therefore \Delta X_t = \gamma X_{t-1} + \sum_{j=1}^{p-1} \delta_j \Delta X_{t-j} + \varepsilon_t$$

- If there is no drift, no trend $\rightarrow$ use above formula for ADF test
- If there is drift only (α_0), then add drift term in ΔX_t

$$\therefore \Delta X_t = \alpha_0 + \gamma X_{t-1} + \sum_{j=1}^{p-1} \delta_j \Delta X_{t-j} + \varepsilon_t$$

- If series contain drift (α_0)+linear trend (βt) $\rightarrow$ add those terms in ΔX_t

$$\therefore \Delta X_t = \alpha_0 + \beta t + \gamma X_{t-1} + \sum_{j=1}^{p-1} \delta_j \Delta X_{t-j} + \varepsilon_t$$

where,

- α_0 = Intercept (drift)
 - β = Coefficient of linear time trend
 - γ = ADF test statistics (testing $H_0 : \gamma = 0 \rightarrow$ Unit root)
 - δ_i = Coefficient of augmented lagged differences.
 - p = Number of lags
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Choosing Lag Order

The number of lagged differences p is usually selected using information criteria such as:

1. Akaike information Criteria (AIC)
2. Bayesian Information Criteria (BIC)
3. Hannan-Quinn Information Criteria (HQIC)
4. Manually using Ng-Perron Formula

1. Akaike Information Criteria (AIC)

Number of lags (p) using AIC,

$$AIC(p) = \ln(\hat{\sigma}_p^2) + \frac{2(P+2)}{n}; \quad \text{where } \hat{\sigma}_p^2 = \frac{\sum_{t=2}^T \hat{\varepsilon}_t^2}{T-2}$$

where,

- P = Number of estimated parameters.
- n = total number of observation
- $\hat{\sigma}_p^2$ = Variance of residuals if number of residuals are ≥ 2

2. Bayesian Information Criteria (BIC)

Number of lag orders using BIC,

$$BIC(p) = \ln(\hat{\sigma}_p^2) + \frac{(P+2) \ln(n)}{n}$$

3. Hannan-Quinn Information Criteria (HQIC)

$$HQIC(p) = -2\hat{L}_{max} + 2P \ln(\ln(n))$$

where, $\hat{L}_{max}$ = Log Maximum Likelihood

4. Ng-Perron Formula

Ng-Perron lag order uses $AIC(p_{max})$ to find maximum number of lags we can obtain.

$$p_{max} = 12 \times \left[\frac{n}{100} \right]^{\frac{1}{4}}$$

where, n = number of total observations

- After finding p_{max} → Run ADF Test → Check if t-state > 1.6
 - if t-state > 1.6 → Stop ADF, current p is optimal lag
 - if t-state < 1.6 → Reduce p_{max} by 1 & repeat the process until 1st condition become true.

Hypotheses

For ADF test, hypotheses remain same as DF test

- $H_0 : \gamma = 0$ → Unit root exist → Series is **Non-stationary**
- $H_1 : \gamma < 0$ → **No Unit Root** exist → Series can be **Stationary**

Test Statistics

The ADF statistic is a **t-statistic** for the estimated coefficient $\hat{\gamma}$:

$$ADF(\tau) = \frac{\hat{\gamma}}{SE(\hat{\gamma})};$$

$$\text{where, } SE(\hat{\gamma}) = \sqrt{\frac{\hat{\sigma}^2}{\sum_{t=2}^T X_{t-1}^2}}; \quad \hat{\sigma}^2 = \frac{\sum_{t=2}^T \hat{\epsilon}_t^2}{T-2}$$

where,

- τ = Critical Value of ADF statistics
- $\hat{\gamma}$ = Estimated Coefficient

- $SE(\hat{\gamma})$ = Standard Error of estimated coefficient

Unlike ordinary regression t-tests, this statistic follows the **Dickey-Fuller distribution**, which is why special critical values are used.

The Dickey-Fuller distribution is **not the standard Student's t-distribution**.

Therefore, the ADF statistic is compared with tabulated critical values.

Critical Values

Confidence Level	Critical Value
1%	-3.50
5%	-2.89
10%	-2.58

Decision Rule:

- If $\tau < \text{Critical Value}$ → Reject H_0 Or Accept H_1 : Series can be **Stationary**
- If $\tau > \text{Critical Value}$ → Fail to reject H_0 Or Accept H_0 : Unit root exist → Series is **Non-Stationary**
- If $\tau = \text{Critical Value}$ → Do KPSS Test

p-Values

p-values are probability of observing $ADF(\tau)$ at least as extreme as the one obtained **assuming the null hypothesis is true**.

- If p-value < 0.05 : Reject H_0 → No unit root exist → Series can be **stationary**
- If p-value ≥ 0.05 : Fail to reject H_0 → Unit root exist → Series is **Non-stationary**